

江模型中的单位线包括了壤中流的坡地出流过程与河网汇流过程，三水源新安江模型的单位线只包括河网汇流，不包括壤中流出流过程。显然，分清水源是比较好的。进一步分析可以看出，不分壤中流的单位线受水源比例的影响比较大，不稳定，参见图 7-4。划分壤中流的单位线只表示河网汇流，不受水源比例的影响，比较稳定。

$$U = \frac{F}{3.6} \times 0.1$$

## 附录C 计算机程序 (PC-1500)

### C.1. 新安江模型

单位线转换系数  
可将径流率转化为流量

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10: DATA 起始值W, WU, WL, WD, QG
20: DATA {雨量P
30: DATA {蒸散发能力EM
40: INPUT "VALUES OF IM, WM, UM, LM, B, C, CG, U,
      K, M, FC", IM, WM, UM, LM, B, C, CG, U, K, M, FC
50: DIM P(M): DIM EM(M)
60: MM = (1 + B) * WM / (1 - IM): MS = (1 + EX) * SM
70: READ W, WU, WL, WD, QG
80: FOR I = 0 TO M
90: READ P(I)
100: NEXT I
110: FOR I = 0 TO M
120: READ EM(I)
130: NEXT I
140: FOR I = 0 TO M
150: PE = P(I) - K * EM(I)
160: IF PE < 0 THEN GOTO 400
170: A = MM * (1 - (1 - W / WM)^(1 / (1 + B)))

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180:IF  $A + PE < MM$  THEN LET

$R = PE - WM + W + WM * (1 - (PE + A)/MM)^{(1+B)}$ :

GOTO 200

190: $R = PE + W - WM$

200:IF  $PE < FC$  THEN LET

$RG = R - IM * PE$ ;  $RS = R - RG$ ; GOTO 340

210: $RG = (R - IM * PE)/PE * FC$ ;  $RS = R - RG$

340: $EU = K * EM(I)$ ;  $D = 0$ ;  $L = 0$

350:IF  $WU + PE - R < UM$  THEN GOTO 380

360:IF  $WU + WL + PE - R - UM > LM$  THEN LET

$WU = UM$ ;  $WL = LM$ ;  $WD = W + PE - R - WU - WL$ ;

GOTO 480

370: $WL = WU + WL + PE - R - UM$ ;  $WU = UM$ ; GOTO 480

380: $WU = WU + PE - R$

390:GOTO 480

400: $R = 0$ ;  $RG = 0$ ;  $RS = 0$

430:IF  $WU + PE > 0$  THEN LET

$EU = K * EM(I)$ ;  $D = 0$ ;  $L = 0$ ;  $WU = WU + PE$ ; GOTO 480

440: $EU = WU + P(I)$ ;  $WU = 0$

450:IF  $WL > C * LM$  THEN LET

$L = (K * EM(I) - EU) * WL/LM$ ;  $WL = WL - L$ ;  $D = 0$ ;

GOTO 480

460:IF  $WL > C * (K * EM(I) - EU)$  THEN LET

$L = C * (K * EM(I) - EU)$ ;  $WL = WL - L$ ;  $D = 0$ ; GOTO

480

470: $L = WL$ ;  $WL = 0$ ;  $D = C * (K * EM(I) - EU) - L$ ;  $WD = WD$   
 $- D$

480: $E = EU + L + D$ ;  $W = WU + WL + WD$

490: $QG = QG + CG + RG * (1 - CG) * U$

500:PRINT USING"###. ##"; R, RS, RG

```

510:PRINT USING "###.##"; QG
520:NEXT I
530:GOTO 80

```

注：如作下列修改就成为三水源新安江模型：

1 ) 把200~210去掉，换上

MS?  
SM?

```

200: X = FR: FR = R/PE: S = X * S/FR: SS = S
210: Q = R/FR: G = INT(Q/5) + 1: Q = Q/G
220: ID = (1 - (1 - (KG + KI))^(1/G))/(1 + KG/KI)
230: GD = ID * KG/KI
235: RS = 0: RG = 0: RI = 0
240: FOR J = 1 TO G
250: AU = MS * (1 - (1 - S/SM)^(1/(1 + EX)))
260: IF AU + Q < MS THEN GOTO 280
270: RS = (Q + S - SM) * FR + RS: GOTO 290
280: RS = (Q - SM + S + SM * (1 - (Q + AU)/MS)^(1 + EX))
      * FR + RS
290: S = J * Q - RS/FR + S
300: RG = S * KG * FR: RI = RG * KI/KG
310: KI = S * ID * FR + RI
320: S = J * Q + SS - (RS + RI + RG)/FR
330: NEXT J

```

2 ) 把400去掉，换上

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400: R = 0: RS = 0
410: RG = S * KG * FR: RI = RG * KI/KG
420: S = S - (RG + RI)/FR

```

3 ) 在10中加上FR, S, 在40中加上KI, KG, SM, EX, 去掉FC.  
在500中加上RI, 在70中加上FR, S.

注：本程序所用标符有下列简化：

IM(IMP), UM(WUM), LM(WLM), CG(KKG), MM(WMM),

SM—自由水 蒸发 计算 过程  
S—自由水 蒸发 计算 过程

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QG(QRG), KI(KSS), MS(SMM), RI(RSS), ID(KSSD),  
GD(KGD), L(EL), D(ED)

### C.2. 陕 北 模 型

```

10: DATA 雨量-蒸发PE, 起始值 I
20: INPUT "VALUES OF F0 FC, K, ER, M", F0, FC, K, ER, M
30: DIM PE(M)
40: FOR J = 0 TO M
50: READ PE(J)
60: NEXT J
70: READ I
80: RT = 0
90: FOR J = 0 TO M
100: T = I/F
110: IB = T * FC - (1 - EXP - (K * T)) * (F0 - FC)/K
120: IF ABS(IB - I) > ER THEN LET
      F = F0 - K * (IB - FC * T): T = T - (1 - IB)/F: GOTO 110
160: F = F0 - K * (1 - FC * T)
170: IF PE(J) > F THEN LET
      R = PE(J) - F: RT = RT + R: GOTO 200
190: F = PE(J): R = 0
200: IF F + I > 50 THEN LET I = 50: GOTO 230
220: I = I + F
230: PRINT USING "###. ###", R, I, F
240: NEXT J
250: PRINT USING "###. ###", RT

```

### C.3. 马斯京根法线性解

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10: DATA 起始流量Q0,,

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